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EDUCATION

The University of Texas at Austin, Austin, TX

Anticipated May 2029

Bachelor of Science in Informatics, Major in Human-Centered Data Science & Cybersecurity

Relevant Coursework: I 310U Introduction to User Experience Design (*in progress*), I 310D Introduction to Human-Centered Data Science (*in progress*), I 305 Research Methods for Informatics, I 303 Ethical Foundations of Informatics, I 306 Statistics for Informatics, AET 304 Foundations of Arts & Entertainment Technology

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#/.NET, Lua, HTML/CSS, POSIX shell

Design & Prototyping: Figma; user research methods, information architecture, competitive analysis; Blender, Onshape, AutoCAD

Frontend: React 18, Next.js, Vite, Tailwind CSS, React Router, HTML/CSS, responsive UI

Backend: FastAPI, Express, Flask, Node.js, Bun, Supabase

AI/ML & CV: XGBoost, scikit-learn, Pandas; YOLO (v8/v11), Roboflow, OpenCV; LLM app development (Gemini, Claude API)

Data/Infra: PostgreSQL, Supabase, Docker, Kubernetes, GitHub Actions, Vercel, Render, Tableau

EXPERIENCE

PERSPECT (tryperspect.com)

Jan 2026 to Present

Founder

- Designed and shipped a solo full-stack AI product where a panel of expert AI personas debates a topic, surfaces disagreements, cites web-searched sources, and outputs a structured briefing; originated at the Claude Hackathon 2026 (top-20 finish)
- Built the frontend on React 18 + Vite + Tailwind + React Router 7 with a multi-persona debate view, contradiction map, synthesized briefing, and follow-up prompts
- Built the backend on Bun + Express with Server-Sent Events streaming for a multi-persona debate pipeline (personas, rebuttals, synthesizer) using Gemini 2.5 with Google Search grounding
- Deployed on Vercel + Render with same-origin API proxy, keep-warm GitHub Action, Gemini spend guard, and CI pipeline (build + dependency audit + secret scan); Supabase for auth and analytics

SBI X UT AUSTIN: DRONE ROAD-INSPECTION RESEARCH, Austin, TX

May 2026 to Present

Tech Lead

- Lead the technical side of a ~10 person research team using drone imagery and computer vision to detect road cracks and stitch images into one continuous georeferenced road map
- Built an end-to-end proof of concept in days: wrote a Python pipeline slicing road-test video into ~785 road-cropped frames, then trained a YOLOv11 crack-detection model locally on GPU at zero cloud cost

BIZFIRST AI, Remote

Jun 2026 to Aug 2026

Software Engineering Intern

- Built and documented BizFirst's V1 self-hosted object storage on MinIO in Docker with tiered quotas scaling to 90% of disk, 365-day lifecycle expiration, and incremental restic snapshots targeting Backblaze B2 for offsite backup
- Wrote a POSIX shell disk-monitoring script with four-tier thresholds (70/85/90/95%) that logs timestamped events and blocks writes before disk exhaustion
- Built three MCP tools (create_form, update_form, delete_form) that let Octopus AI, BizFirst's multi-agent orchestrator, generate valid Atlas Forms JSON schemas and call the REST API from natural-language prompts
- Authored a storage-platform architecture analysis tracing the redesign of a 493-line single-file prototype into a two-module Clean Architecture system, documenting 12 design patterns, AES-256-GCM envelope decryption, and TTL credential caching with thundering-herd protection
- Contributed to StorageStudio (open-source React UI for S3-compatible storage, owning backend API design) and AccumulateNodes (Accumulate blockchain integration in C#/.NET, alongside Factom founder Paul Snow)

SUSTAINABLE BUILDING INITIATIVE (UT AUSTIN), Austin, TX

Aug 2026 to Present

Tech Director

- Leads a 10 to 15 person tech team building the organization website and related technical projects, reviewing and approving all work before branches merge to main
- Serves as primary technical point of contact for client-facing communication and runs recruiting, interviews, and onboarding for new tech team members
- Leads the External Technologies function, tracking industry research and emerging tooling to inform the team's tech-stack decisions

SUSTAINABLE BUILDING INITIATIVE, Austin, TX

Feb 2026 to Aug 2026

Software Developer, Tech Team

- Shipped features for the SBI organization website on a 5 to 7 person tech team, meeting twice weekly to hit deliverables on time
- Designed and shipped the public-facing Reports feature end to end using Supabase (Postgres + storage bucket with image validation), Next.js, and Bun
- Integrated the front end with the Supabase backend for a production-ready reports workflow

HC4A: AI WITH ALL HACKATHON, Austin, TX

Jul 2026 to Present

Tech Lead

- Co-leads technology and student-council operations for Hindu Charities for America's hackathon (~600 participants, grades 8-12), coordinating with a data scientist and an AI/UX designer on a one-month build
- Designed the app's screens in Figma: a live "Now / Next" status ticker to cut cognitive load during a high-velocity event, a preset User ID sign-in to remove account-creation drop-off, a self-guided design-sprint roadmap, and a faceted directory across nonprofits, sponsors, and schools; benchmarked against SXSW, MLH, and Devpost
- Designs the participant app feature set and a privacy-aware per-participant UID scheme under COPPA and FERPA constraints, scoped for 500 students, 100 teams, and 20 pods across two venues

PROJECTS

NETSENTINEL: ML-powered Network Threat Detection

2026

- Shipped a FastAPI backend with WebSocket threat broadcasts and a React 3D globe dashboard; deployed via Docker Compose and Kubernetes manifests
- Python, FastAPI, XGBoost, React, TypeScript, Kubernetes, Docker* (github.com/Nkbros12k/netsentinel)

GRAVITY-ZERO: Fully Local AI IDE

2026

- Built a fully local AI IDE running a three-agent triad (Planner, Coder, Reviewer) over a quantized Qwen 2.5 Coder 7B model served by LM Studio; no code or data leaves the machine
- Shipped a Monaco-based editor with a FastAPI backend coordinating the agent handoffs
- Python, FastAPI, React, LM Studio* (github.com/Nkbros12k/gravity-zero)

Leadership & Honors: Founder, Blood Buddy, a consumer product with 8,000+ donors; National Cyber Scholar (2024); CyberPatriot State Semi-Finalist; SANS Foundations + GIAC.